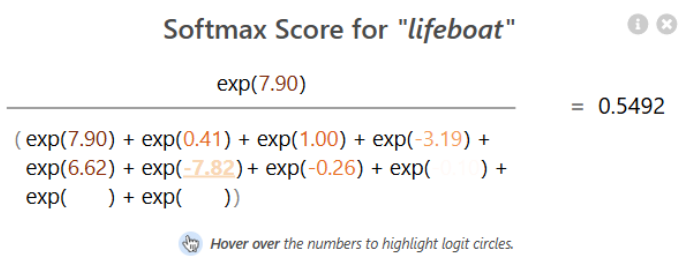
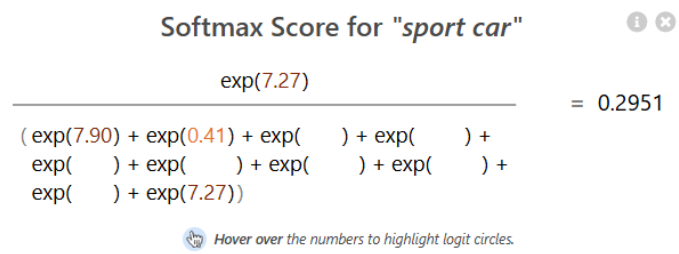
CNN Explainer activity - uploaded images

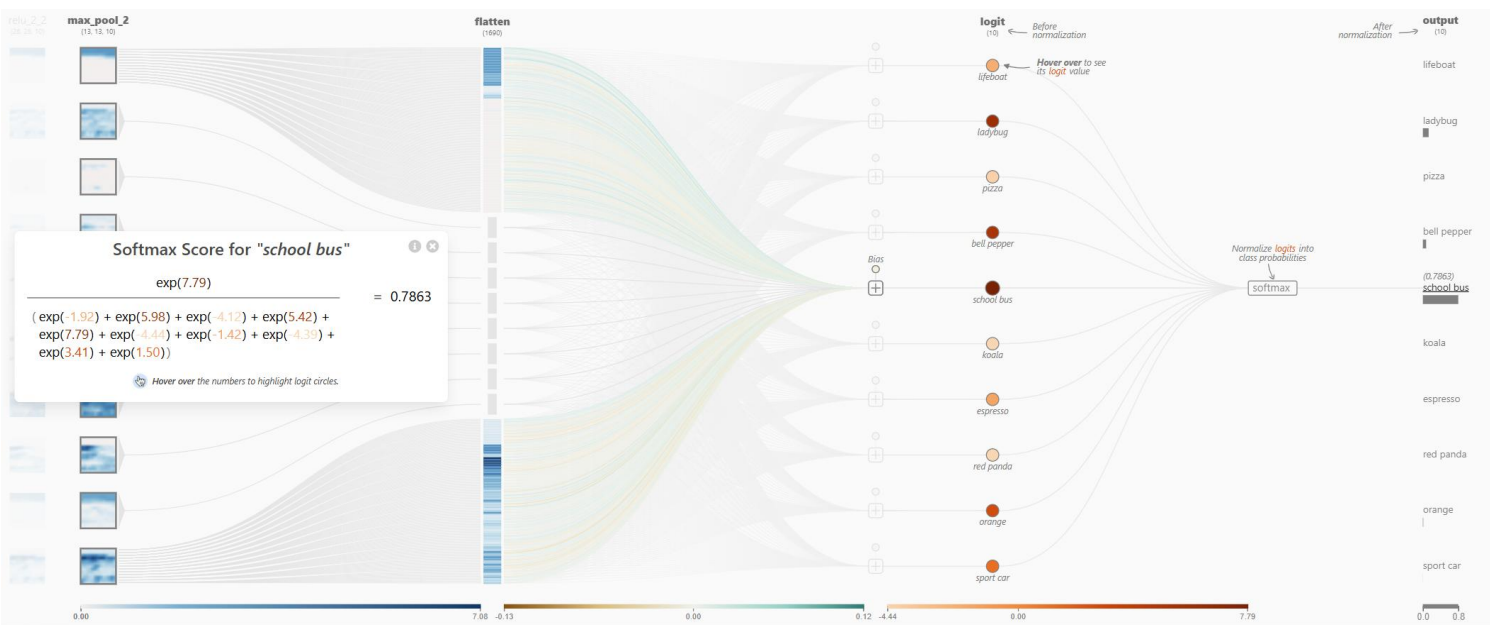
1. Uploaded a sports car image into CNN Explainer to test how the CNN classifies a real-world example.

- Model output (softmax probabilities):
    - lifeboat: 0.5492 (top prediction)
    - sport car: 0.2951
    - school bus: 0.1538





- Interpretation:
    - The model misclassified the sports car as “lifeboat”, but still assigned a substantial probability to “sport car”.
    - This suggests the learned features extracted from the cropped and resized input triggered patterns that aligned more strongly with the lifeboat class for this specific image.
- Layer observation:
    - Feature maps became more abstract and localised in later convolution layers, and pooling reduced spatial detail while keeping the strongest activations feeding into the final logits and softmax output.
- Evidence captured:
    - Screenshot of the uploaded image and the output probability panel to document the top labels and confidence split.

2. Uploaded a landscape image that is outside the model's label set to test how the CNN behaves on out-of-distribution inputs.

- Model output (softmax probability):
  - school bus: **0.7863** (top prediction)
- Interpretation:
  - Even though the image is not a school bus, the model still assigned a confident label because it must choose from a fixed set of classes.
  - This demonstrates a key limitation of closed-set classification, CNNs can appear confident on unfamiliar inputs if the visual patterns partially match features learned for a trained class.
- Evidence captured: